

CSE/DSC 234 Project 2 – Schema Linking with SFT

Final Report

Final validation leaderboard: 0.7270 (Table 0.7911 / Column 0.6630). Baseline (smoke adapter): 0.5605. Best single adapter: 0.6648 (sweep 13). Submission: `python3 main.py --input X --output Y` runs a **4-way LoRA ensemble with majority-2 aggregation across two base models AND mixed retrievers per adapter**: sweep 22 on Qwen3-1.7B + embed, sweep 13 on Qwen-Coder-1.5B + embed, sweep 15 on Qwen-Coder-1.5B + BM25, and sweep 13 *reused* on Qwen-Coder-1.5B with hybrid retrieval. An identifier survives iff it is emitted by ≥ 2 of the 4 passes. Completes 101 questions in **13m9s** on a 24 GB MIG slice (<15 min budget). `main.py` reads each adapter’s `adapter_config.json` to group adapters by base model (so each base is loaded only once) and accepts a `path[:retriever]` spec per adapter, so the same adapter under different retrievers can be ensembled too. The 4th pass is exactly that – pure post-hoc diversity, no extra training.

1 Training Data Methodology

The released 301 train + 101 validation split was the starting point. We treated schema linking as next-token prediction over a chat prompt: a fixed system message explains the JSON contract and wildcard convention; the user turn carries the serialized schema and NL question; the assistant turn carries the canonical JSON target (sorted keys + sorted columns) emitted by `prompt.target_string`.

Schema serialization. The baseline serializer is one line per table, `TableName: c1, c2, ...`. We tested two richer variants: (i) **types** – appending column type in parentheses; (ii) **keys** – * for primary keys and `->TargetTable.col` for foreign keys. A combined **types_keys** serialization roughly doubled token counts on the dense SBO schemas; on the largest (SBO-Banking) the prompt exceeded 6k tokens. **keys** alone added no tokens on SBO because the SBO Spider schemas carry no PK/FK metadata – only 5 of the 17 schemas do (ATBI, NTSB, NorthernPlainsFireManagement, KlamathInvasiveSpecies, PacificIslandLandbirds).

Output format. Raw JSON with stable casing, alphabetical keys and column lists. A table with no specific columns (wildcard queries like `count(*) from t`) is emitted as `{"t": []}`, not omitted. We verified this end-to-end with `tests/test_checklist.py`; only 3 validation questions have empty-gold tables but failing to emit them is a recall miss on both axes.

Table-level retrieval at training time. Schemas with > 500 columns (all 9 SBO modules + NTSB + NYSED) get BM25-filtered to the top-20 tables. **Oracle augmentation:** at training time we force-include every gold-referenced table in the kept set so the model is never asked to predict labels absent from its prompt. Inference does not have this oracle.

Augmentation. The 9 SBO modules ship with only 6–12 train examples each (~22% of train) versus 24–60 for the parks/NTSB domains; SBO is also where ~55% of our validation table-misses concentrate. We generated 216 additional (question, gold_sql) pairs via the Claude API (Haiku 4.5; 9 batches of 24) focused entirely on SBO, labeled each with the project’s `data_prep/sql_to_schema_links.py` (sqlglot-based) extractor, and dropped any that failed to parse or referenced 0 schema tables. Only 1 of 217 candidates was dropped (a duplicate question; 0 hallucinated tables, 0 parse errors – evidence the prompt and validation pipeline are tight). The merged `train_augmented.json` has 517 examples. The audit log lives at `data/train_augmented_audit.json`.

2 Final Pipeline Architecture

Base model. Qwen/Qwen2.5-Coder-1.5B-Instruct. The code-pretrained variant of Qwen2.5-1.5B; same parameter count, same ChatML template, same tokenizer; SQL/schema pretraining produced measurable gains over the vanilla Instruct variant (Sec. 4).

Adapter. LoRA $r=32$, $\alpha=64$, dropout 0.05, target modules `{q,k,v,o,gate,up,down}_proj` (the full attention block plus the MLP block). bf16 with gradient checkpointing during training. Adapter safetensors are shipped as **fp16** (PEFT’s default fp32 save was 148 MB per adapter, above GitHub’s 100 MB per-file limit; fp16 halves that to 74 MB and we verified the score is preserved within ± 0.001 of fp32). The 3 adapters together fit in ~220 MB of repo size.

Optimizer. AdamW (TRL default), `lr_scheduler_type=linear`, `warmup_ratio=0.05`, `per_device_train_batch_size=1`, `gradient_accumulation_steps=4`, `max_steps=200`, `max_length=4096`. Three adapters are shipped: sweep 13 (Qwen-Coder-1.5B, $lr=3e-4$, 301 train), sweep 15 (Qwen-Coder-1.5B, $lr=3e-4$ + augmented data, 400 steps), and sweep 22

(Qwen3-1.7B, lr=3e-4, 301 train). The 4th inference pass reuses sweep 13’s weights with a different retriever – so the 4-way ensemble runs over 3 trained adapters, not 4.

Inference pipeline (main.py). Adapters are grouped by their declared base model (read from `adapter_config.json`); each base loads once. Within a group, adapters are loaded as named PEFT adapters and swapped via `model.set_adapter`. For each (adapter, retriever) pass we accumulate per-question {table→cols} predictions and per-pass vote counts; after the last pass we apply `--aggregation maj2` (default; `union` also supported). **Partial-save safety contract:** passes 1–3 write the running union, pass 4 writes `maj2`. Combined with the deliberate adapter order (Qwen3 first), this means: after 3 passes the on-disk file matches the previous locked champion (0.7107); a mid-pass-4 kill ships that baseline; clean completion ships 0.7270. **Retrieval:** BAAI/bge-small-en-v1.5 cosine similarity (top-20 tables) for passes 1–2, BM25 for pass 3 (sweep 15), RRF hybrid for pass 4 (sweep 13). The cross-retriever diversity is what `maj2` converts into score.

Post-processing. `canonicalize_prediction` drops any non-schema identifier (precision-preserving hallucination filter) and re-cases identifiers to schema casing. The final submitted output had 0 schema-invalid identifiers across all 101 validation predictions.

3 Experimentation Methodology

We ran 20 numbered sweeps via **RapidFire AI** (`training/train_rapidfire.py`, `RFGGridSearch/RFSFTConfig`). Sweeps 1–9 were spent surviving infrastructure problems (described below); sweeps 10–20 produced the substantive results in Table 1. Per-sweep `rapidfire.log`, `training.log`, and extracted MLflow metrics (`metrics.json`) live under `logs/sweepN/`; full per-run configs are pulled from the rapidfireai MLflow store on demand.

3.1 Sweep dimensions

Across the three *training-time* axes – base model, adapter capacity, training data – and four *inference-time* axes – prompt style, retriever, ensemble strategy, post-processing – we ran 11 distinct training configs and ~10 distinct inference-time variants. Configs were chosen one-at-a-time A/B style off the current champion so each result is attributable to a single knob. Multi-config sweeps (8–16 in `RFList`) were used in our first attempts (sweeps 5–8) but, after infrastructure issues forced us to `num_chunks=1` (Sec. 3.2), we switched to single-config sweeps and let the cost of more sweeps buy clean attribution.

3.2 Infrastructure issues and how we worked around them

Issue #1: DSMLP cleanup SIGKILLs (~24h of lost GPU time). Sweeps 5–8 ran multi-config `RFGGridSearch` with 8–16 configs and `max_steps=300`. Each tried to train through ~5–6h of wall time. DSMLP enforces a process-cleanup window we did not anticipate – somewhere around the 4–6h mark, our worker process got SIGKILL’d with no warning. The mlflow metrics survived (they are written to a SQLite DB on `$HOME`) so we could see that sweep 8 had reached step 264/300 across all 8 configs before dying. But the trained LoRA weights themselves had *nowhere on disk*. We initially assumed the problem was the kill itself; the real story turned out to be deeper.

Issue #2: RapidFire’s SHM-only checkpoint store. While debugging the loss, we ran sweep 9 with only 2 configs and `max_steps=200`, expecting it to finish in well under the kill window. It did – “Experiment ended” fired cleanly, no SIGKILL. *We still had no final_checkpoint/*.safetensors on disk.* Reading `rapidfireai/fit/backend/worker.py` around L440 revealed why. By default RapidFire writes intermediate checkpoints to `/dev/shm` via its `shm_manager`; the disk write only fires inside a single branch gated on `is_run_finished == (chunk_id == num_chunks-1) && (steps ≥ total_steps)`. With the controller’s round-robin chunk schedule, a run typically hits its step budget on chunk N where $N < \text{num_chunks}-1$, the disk-save branch is skipped, and the in-SHM adapter dies with the worker process at `experiment.end()`. We considered patching the constant `USE_SHARED_MEMORY = True` at module level but rejected that as fragile (it affects all rapidfireai usage on the system and would not survive a `pip install --upgrade`). The clean workaround: pass `--num_chunks 1` so every run necessarily ends on “the last chunk” by construction and the disk-save branch fires reliably. All sweeps from 10 onward use this workaround; sweep 10 was the first successful end-to-end persisted RapidFire training in our series.

Issue #3: HuggingFace Trainer fallback (training/train_single.py). While debugging Issue #2, we were not sure whether the problem was orchestration (rapidfireai) or training (TRL/PEFT setup). We built `training/train_single.py` – a ~120-line plain `trl.SFTTrainer` script – as an A/B against the same config. It trained fine in 14 minutes and produced a 0.6078 adapter, confirming that the training stack worked end-to-end and isolating the failure to the rapidfireai chunked-orchestration path. After we landed Issue #2’s fix this script became dead code, but it stayed in the repo as a stress-tested single-config fallback.

Issue #4: CUDA SIGFPE in main.py at batch_size>1. On the first ~ 0.61 inference run we hit a hard “Floating point exception (core dumped)” right at the first `model.generate` call. Process crashed before producing any predictions. We narrowed the trigger by bisecting the CLI flags and found that batch size 1 worked, ≥ 2 SIGFPE’d. Stack combination is `transformers 4.57 + torch 2.10 + bf16 + left-padded prompts  $\sim 4k$  tokens + H100`. We did not find a root-cause issue but the pattern matches reports of a kernel edge case in the attention path with long left-padded sequences in bf16. We defaulted `batch_size=1` in `main.py` (with a comment) and noted it in the README so a grader doesn’t trip over it.

Issue #5: Mismatched val-data layout. Halfway through development we reorganized the repo to push raw data under `data/`; the `tests/test_checklist.py` script (the rubric quick-start verifier) was written against the old root-level paths. We updated the test, kept it green, and added it to the development loop – it’s a cheap sanity check that every commit doesn’t break the I/O contract (per-file question_ids, schema-folder layout, wildcard-table preservation, hallucination filtering).

3.3 The supporting tools we built

`data_prep/` holds data-side scripts: `augment_data.py` (SBO augmentation via Haiku 4.5, SQL validated through `sql.to_schema.links.py`, audit log preserved), `format_training_data.py` / `format_two_stage_data.py` (JSON $\rightarrow$ JSONL), `generate_cot_traces.py` (sweep 21 rationales), and `expand_columns.py` (Sec. 4 post-processor). `scripts/probe_ensembles.py` is the combinatorial probe that discovered the 4-way + maj2 (138 cells, calls `eval.py` on each, ranks by score). `logs/README.md` maps each sweep to its config and metrics from the rapidfireai MLflow store.

3.4 Inference-time experimentation

Once we had the single-best adapter (sweep 13), we tested eight inference-time variants: `max_tables 20 $\rightarrow$ 40`, three retrievers (BM25 / embed / hybrid via RRF), self-consistency at $K=3$ (union-of-samples and majority-vote), column-expansion post-processing, union/intersection/ majority-k ensembling over progressively more adapters (2–6 way), and *mixed-retriever passes over the same adapter* (sweep 13 once with embed and once with hybrid as separate ensemble members). Variants were A/B’d directly against the previous champion file in `predictions/` – no retraining, ~ 5 –15 min per A/B; the 4-way + maj2 search was a combinatorial offline probe over 138 (combo, aggregation) cells against the per-(adapter, retriever) prediction files we had accumulated. Many variants were negative (Sec. 4) but two stood out: embedding retrieval (+0.014 single, +0.016 in the ensemble), and the 4-way + maj2 combination (+0.016 over the 3-way mixed-retriever).

4 Results and Analysis

What worked.

- **Switching to Coder-1.5B (+0.003):** small but in the right direction; sets up the bigger gains.
- **Bigger LoRA: $r=32$ + MLP modules (+0.039):** the single biggest training-time lever. The full attention+MLP adapter has $\sim 3.5\times$ more trainable parameters than $r=16$ q/k/v/o and converged to lower train loss (0.075 vs 0.108) without overfitting at 200 steps. Notably this is the configuration our sweeps 5–7 attempted but lost to the SHM bug – the `num_chunks=1` workaround was load-bearing for actually *getting* this number.
- **Embedding-based table retrieval (+0.014 single, +0.016 ensemble):** BAAI/bge-small-en-v1.5 surfaces semantically related tables that BM25 misses (e.g. qid 95 needs ORCT, ranked outside BM25 top-10 because all top hits are PWZ* payment-wizard children; the embedder puts it at rank 1). The biggest *inference-time* lever.
- **Union ensemble of 3 diverse adapters (+0.034 over best single):** sweep 13 ($lr=3e-4$, 301 train), sweep 15 ($lr=3e-4$, 517 train, 400 steps), and sweep 18 ($lr=2e-4$, 301 train). Chosen because their errors are sufficiently uncorrelated; 4- and 5-way variants regressed because the weaker fourth model (sweep 20 / sweep 12) brings more noise than signal.
- **Adding Qwen3-1.7B (sweep 22) for cross-family ensemble (+0.017 over single-base champion):** sweep 22 alone scores 0.66 – weaker than sweep 13 (0.66) – but its errors are markedly less correlated with the Qwen-Coder ensemble members because it comes from a different model family entirely. Replacing sweep 18 (the weakest Coder member) with sweep 22 produced the 3-way 13+15+22 all-embed ensemble at 0.7019. The submission’s `main.py` groups adapters by base model and loads each base only once, so this multi-base ensemble still fits in the 15-minute inference budget.
- **Mixed-retriever ensemble (+0.009 over the all-embed champion):** the same adapter produces different predictions under different retrievers because the candidate-table set in the prompt differs. Putting sweep 15 under BM25 (while 13 and 22 stay on embed) gave us 0.7107 – the right retriever for each ensemble member is whichever

#	base model	key knobs	lr	steps	val LB	Δ
10	Qwen2.5-1.5B	r=16 q/k/v/o, BM25	3e-4	200	0.6229	—
12	Qwen2.5-Coder-1.5B	r=16 q/k/v/o, BM25	3e-4	200	0.6258	+0.003
13	Qwen2.5-Coder-1.5B	r=32 + MLP, BM25	3e-4	200	0.6648	+0.039
18	Qwen2.5-Coder-1.5B	r=32 + MLP, BM25	2e-4	200	0.6609	-0.004
19	Qwen2.5-Coder-1.5B	r=32 + MLP, BM25, cosine LR	3e-4	200	0.6479	-0.017
14	Qwen2.5-Coder-1.5B	r=32 + MLP, BM25, 517 train (aug)	3e-4	200	0.6379	-0.027
15	Qwen2.5-Coder-1.5B	r=32 + MLP, BM25, 517 train, 400 steps	3e-4	400	0.6600	-0.005
16	Qwen2.5-Coder-1.5B	r=32 + MLP, BM25, 301 train, 400 steps	3e-4	400	0.6268	-0.038
11	Qwen2.5-1.5B	r=16 q/k/v/o, types_keys prompt	3e-4	200	0.5946	-0.028
20	SmolLM2-1.7B	r=32 + MLP, BM25	3e-4	200	0.5301	-0.135
17a/b	Qwen2.5-Coder-1.5B	Two-stage: table model + column model	3e-4	200	0.6127	-0.052
21	Qwen2.5-Coder-1.5B	r=32 + MLP, CoT-distilled training data	3e-4	200	0.6103	-0.054
22	Qwen3-1.7B	r=32 + MLP, BM25, original data	3e-4	200	0.6599	-0.049
27	Qwen2.5-Coder-1.5B	r=32 + MLP, embed-prompted train	3e-4	200	0.6772	-0.001 vs s13 embed
28	Qwen2.5-Coder-1.5B	r=32 + MLP, keys prompt (PK/FK markers)	3e-4	200	0.6464	-0.032
Inference-time – best single adapter (sweep 13):						
BM25 retrieval (baseline)						0.6648
max_tables: 20 → 40 (more distractors)						0.6032
Embedding retrieval (BAAI/bge-small-en-v1.5)						0.6786
Hybrid retrieval (RRF of BM25 + embed)						0.6724
Self-consistency, K=3, T=0.5, union-of-samples						0.6737
Column expansion (question-keyword ↔ col-name match)						~same
Ensembles – 3-way Coder-only (sweeps 13+15+18):						
BM25 retrieval						0.6986
Embedding retrieval (single-base champion before Qwen3)						0.7002
Hybrid retrieval (RRF)						0.6977
2-way 13+18, 4-way +14 / +20 / +11						0.66–0.69
Ensembles – cross-family (Coder + Qwen3):						
2-way 13+22 (both embed)						0.7063
3-way 13+15+22 all-embed (single-retriever)						0.7019
3-way mixed-retriever: 13.embed + 15.BM25 + 22.embed (previous champion)						0.7107
4-way + maj2: above three + 13.hybrid, ≥2-vote aggregation						
(submitted champion: same adapters, no extra training; 13m9s wall time)						0.7270
same 4-way under partial-union aggregation (regression – shows maj2 is doing the work)						0.7034

Table 1: All 13 distinct training configs (top) and the inference-time ablations on the best single adapter and the 3-way ensemble (bottom). Bold marks the single best config in each section. Δ is vs sweep 10 (the single-config baseline). r=32 + MLP means LoRA r=32 with target modules {q,k,v,o,gate,up,down}_proj. BM25 retrieval scores tables by query-vs-(table+cols) BM25; embed replaces BM25 with sentence-transformer cosine similarity (BAAI/bge-small-en-v1.5); hybrid is Reciprocal Rank Fusion of the two.

decorrelates its errors from the others, not the one used at training time. Exposed in main.py’s --ensemble_dirs via a path[:retriever] spec per adapter.

- **4-way + majority-2 aggregation (+0.016):** a combinatorial offline probe over our accumulated per-(adapter, retriever) prediction files found that adding a 4th pass – sweep 13 reused under hybrid retrieval, no extra training – and switching aggregation from union to **majority-2** (≥2 of 4 votes) lifts the leaderboard to **0.7270**. The same 4-way under union actually regresses to 0.7034: maj2 is doing the work, not the extra pass. Mechanically this is the precision/recall trade we needed – union maximizes recall (table F1 was already 0.79); maj2 prunes column-level false positives. Both axes improved (column 0.6423→0.6630, table 0.7791→0.7911). Wall time 13m9s, inside the 15-min budget.

What didn’t work, and why.

- **types_keys prompt (-0.028):** roughly doubled token count and *regressed* both table and column scores. We attribute this to attention dilution – with ~2× more schema text per prompt, the 1.5B model focuses less precisely on the 2–4 actually-relevant tables out of 20. Cheap signal at this scale beats expensive signal.
- **max_tables=40 (-0.020):** contrary to our prior that exposing more candidate tables would recover BM25-cut gold tables. Distractor tables hurt the model’s table precision more than the marginal recall recovery.

- **Two-stage (table model then column model) (-0.052)**: a clean architectural negative result. Stage-A loses the column signal that disambiguates similarly-named SBO siblings (OPMG vs OPHA); stage-B cannot reason cross-table about JOIN columns; errors compound (a wrong table in stage-A can't be corrected by stage-B). Joint reasoning is load-bearing for this task.
- **SmolLM2-1.7B base swap (sweep 20, -0.135 alone)**: the model is genuinely weaker on this distribution (token accuracy 0.80 vs Coder's 0.90); adding it to the ensemble *hurt* (-0.037), so diversity does not always help when the noise floor of the new member is too high.
- **Data augmentation alone (sweep 14, -0.027)**: train_loss $\sim 2\times$ higher than the 301-example baseline at 200 steps (underfitting); doubling steps (sweep 15) recovered most of the regression but did not exceed the baseline. The 2×2 table-size/step-count grid implies the augmented data is roughly neutral here – the limiting resource is per-example exposure, not raw data quantity.
- **Self-consistency union-of-samples (-0.005) and column-expansion post-processing (neutral)**: $T=0.5$ sampling adds noise; question-keyword $\leftrightarrow$ col-name match either over-fires (>1 token: -0.009) or under-fires (>2 meaningful tokens: only ~ 9 cols, wash). Greedy decode + the precision-preserving canonicaliser do the work already.
- **Chain-of-thought distillation (sweep 21, -0.054)**: brief (~ 50 token) Haiku-generated rationales were schema-grounded (300/301 referenced at least one gold table) but the CoT-trained adapter landed at ~ 0.61 and regressed the ensemble by 0.007. Likely a mix of: gradient budget spent on reasoning tokens, descriptive rather than disambiguating rationales, and early commitment that propagates errors into the JSON.
- **Embed-prompted training (sweep 27, neutral)**: sweep 13's recipe with the training-time table filter switched from BM25 to embed. The prior was that closing the small train/test retrieval gap would yield $+0.005$ – 0.02 ; standalone the adapter scores 0.6772 (vs sweep 13's 0.6786 under embed inference – identical), and swapping it into the 4-way lifts LB $+0.002$ entirely on the table side (column F1 $0.6630 \rightarrow 0.6631$). Retrieval mismatch is not the column-F1 bottleneck.
- **Keys prompt style (sweep 28, -0.032 alone, regresses every ensemble swap by 0.008–0.028)**: trained with PK * markers and `->TargetTable.col FK` arrows, expecting it to disambiguate FK-shared columns (CASEID across NTSB tables – the top column-F1 FP class in our error analysis). The FK metadata annotates *boring* columns (VERSION->CHILDSEAT.VERSION); the actually-confusing ones (CASEID, EVENT_ID, VEHNO) appear without markers because the schema doesn't list them as FKs. The prompt added structure for non-problems.

Which knobs mattered for table vs column F1. The submitted 4-way's table score (0.7911) is 13 points above its column score (0.6630), and this asymmetry is consistent across all our adapters and across the iterative champions (3-way locked: 0.7648 / 0.6356; 4-way maj2: 0.7911 / 0.6630). Sweep 13's $r=32$ +MLP upgrade lifted column F1 the most among training changes ($+0.045$ vs sweep 12); embedding retrieval lifted column recall ($+0.149$ over BM25 in the 3-way ensemble). The maj2 switch was the first inference-time lever that improved *both* axes (table $+0.012$, column $+0.021$): it shaves the union's column-level false positives without losing the recall that drove the 3-way's table side. Going the other way, the SmolLM2 swap collapsed both axes evenly – suggesting column performance is bottlenecked by base-model capacity and retrieval quality, not by prompt or post-processing.

What we would try next. (1) A column-specialist second stage conditioned on the 4-way's table predictions (which are already 0.79); sweep 17's two-stage failed because stage-A had nontrivial table noise – conditioning on ensemble-agreed tables changes that failure mode. (2) A bigger base for cross-family ensembling (Coder-3B); sweeps 27 and 28 suggest the column-F1 ceiling around 0.66 is set by base capacity, not prompt or data. (3) Hybrid retrieval with a learned cross-encoder reranker over BM25 $\cup$ embed top-30 candidates.

5 AI Tools Statement

Code was developed iteratively with Anthropic's **Claude Code (Claude Opus 4.7)**; the SBO augmentation and CoT-trace pipelines call the Anthropic API using **Claude Haiku 4.5**. All design decisions, sweep selection, eval interpretation, and final configuration choices were ours. The audit log `data/train_augmented_audit.json` records every generated example and its validation outcome for transparency.